



Collision between soliton and polarization domain walls in fiber lasers

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Abstract: We report on the experimental observation of coexistence of solitons and polarization domains, as well as features of soliton collision with polarization domain walls (or kink solitons) in a single mode fiber laser. Depending on the sign of cavity dispersion, either bright or dark solitons have been formed in our fiber laser. Under suitable laser operation conditions, they could even coexist with polarization domains and collide with the domain walls. We show experimental evidence of both the elastic and inelastic soliton collisions with the domain wall solitons. Some interesting features of soliton interaction with polarization domains in a fiber laser are also firstly revealed.

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1. Introduction

Solitons are the hallmark of nonlinear physics. They are ubiquitous in nature and appear in diverse systems ranging from water waves [1], carbon nanotubes [2], Bose-Einstein condensates [3] and optical fibers [4]. In nonlinear optics, soliton propagation in single mode fibers (SMFs) has highlighted the rich manifold of soliton dynamics [5]. Light propagation in single mode fibers is well governed by the nonlinear Schrödinger equation (NLSE), in which soliton formation is a consequence of the natural balance between the fiber dispersion and the nonlinear self-phase modulation [6,7].

Compared with the scalar NLSE, coupled NLSEs admit richer nonlinear wave dynamics. In the framework of light propagation in birefringent fibers, considering the cross-phase modulation (XPM) effect between the two orthogonally polarized light beams, various new phenomena such as polarization switching [8], polarization modulation instability [9], coherently / incoherently coupled vector solitons [10–13], and polarization domains [14–16] could be produced. Among these phenomena, formation of vector solitons and polarization domain wall solitons has attracted considerable attention. Here vector solitons are meant the solitons that have two coupled orthogonal polarized components. Theoretically, various vector solitons, such as the bright-bright [10], dark-dark [4], and dark-bright form [12,13], were predicted. These vector solitons were also experimentally confirmed [17–20]. The polarization domains formed between two counter-propagating beams in SMFs was first theoretically predicted by Wabnitz [21]. It was later experimentally confirmed by Pitois *et al.* [22]. Very recently, Tang *et al.* [15] and Lecaplain *et al.* [14] have further shown both theoretically and experimentally that polarization domains could even be formed in fiber lasers with a weakly birefringent cavity. Associated with the formation of polarization domains, the domain walls that separate the polarization domains are proven to be a stable localized structure. They constitute a fundamentally new type of solitary waves, known as polarization domain wall solitons (PDWS) or topological solitons [23]. Polarization domain wall solitons also have been experimentally confirmed [24,25]. Formation of optical domains and domain wall solitons is a general phenomenon of the coupled nonlinear systems where multiple stable states coexist in the same parameter regime [26]. Indeed, not only domain wall solitons

between two orthogonal polarizations, but also domain wall solitons between two or multiple lasing wavelengths have been revealed in fiber lasers [27,28].

However, majority of previous studies have mainly focused on the features of each individual phenomenon of the soliton or the optical domain formation. The coexistence of them and the features of their mutual interactions are less addressed. Given the case of light propagation in a birefringent SMF, it is to expect that under suitable conditions, either the scalar or the vector solitons could coexist with the polarization domains (or PDWS) in the system. How would they interact and affect each other? Some previous studies have investigated collision of the NLSE solitons either in spatial [29,30] or temporal case [31–35]. However, to our knowledge, no experimental studies on collision between soliton and polarization domain wall (or PDWS) are reported.

In this paper, we report on our experimental results of soliton collision with polarization domain walls (or PDWS). We have designed and operated an erbium-doped fiber ring laser with weak cavity birefringence either in the net normal or net anomalous cavity dispersion regime, and successfully generated a state of coexistence of bright (or dark) solitons with polarization domains in the laser cavity. Based on such a laser system we then experimentally investigated the collision of the solitons with the polarization domain walls and revealed the existence of both elastic and inelastic collisions between them.

2. Experimental setup

The light propagation in a weakly birefringent single mode fiber is described by the coupled NLSEs as follows,

$$\begin{aligned}\frac{\partial u}{\partial z} &= i\beta u - \delta \frac{\partial u}{\partial t} - i\frac{\beta_2}{2} \frac{\partial^2 u}{\partial t^2} + i\gamma \left(|u|^2 + \frac{2}{3}|v|^2 \right) u + i\frac{\gamma}{3} v^2 u^* \\ \frac{\partial v}{\partial z} &= -i\beta v + \delta \frac{\partial v}{\partial t} - i\frac{\beta_2}{2} \frac{\partial^2 v}{\partial t^2} + i\gamma \left(|v|^2 + \frac{2}{3}|u|^2 \right) v + i\frac{\gamma}{3} u^2 v^*\end{aligned}\quad (1)$$

Where u and v are the two normalized slowly varying amplitude envelopes along the orthogonally polarized axes, $2\beta = 2\Delta n/\lambda$ is the wavenumber difference, and $2\delta = 2\beta\lambda/2\pi c$ is the group velocity difference, β_2 is the group velocity dispersion (GVD) coefficient, and γ is the nonlinear coefficient of the fiber.

Previous studies have shown that under suitable conditions the soliton formation and dynamics in a fiber laser could be well described by the NLSE or coupled NLSEs [17–20]. In a previous paper we reported polarization domain formation in a net anomalous dispersion cavity fiber laser [15] and have showed that its feature could be understood based on the coupled NLSEs. Automatic scalar and vector soliton formation in fiber lasers even without any saturable absorbers in cavity was also experimentally observed. It has been shown that the automatic soliton formation in such fiber lasers is related to the various laser instabilities [36]. However, to obtain laser instability a certain appropriate design on the effective cavity dispersion and birefringence is needed.

In our current experiment we adopted the cavity dispersion- and birefringence-management technique [20] to control the cavity properties. A schematic of our fiber laser is shown in Fig. 1. It has a fiber ring cavity made of 3 m Erbium doped fiber (EDF) with a normal group-velocity dispersion (GVD) coefficient of $61.18 \text{ ps}^2/\text{km}$, and different lengths of dispersion-compensating fiber (DCF) with GVD coefficient of $5.1 \text{ ps}^2/\text{km}$, and single mode fiber (SMF) with GVD coefficient of $-22.94 \text{ ps}^2/\text{km}$. Both the SMF and DCF fibers are used because through selecting appropriate lengths, the desired average cavity GVD as well as the cavity length could be obtained. The fiber laser is pumped by a SMF Raman laser operating at 1480 nm. The pump laser has a maximum output power of ~5W. A polarization-independent isolator is inserted in the cavity to force the unidirectional operation of the ring, and an intra-cavity polarization controller (PC) is

used to fine-tune the linear cavity birefringence. A wavelength division multiplexer (WDM) is used to couple the pump light into the cavity, and a 10% fiber coupler is used to output the laser emission. All the components used in cavity are pigtailed with the SMF. No real or artificial saturable absorbers are in the cavity. Outside of the cavity, a pigtailed polarization beam splitter is used to separate the two orthogonal polarization components of the laser emission. The output is monitored with a high-speed electronic detection system comprising two 40-GHz photodetectors and a 33G-GHz bandwidth real-time oscilloscope. An optical spectrum analyzer is used in our experiment to monitor the optical spectrum of the laser emission.

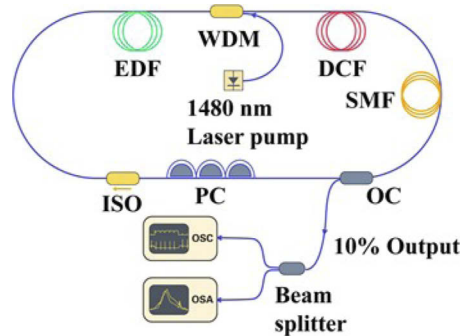


Fig. 1. Schematic of the fiber laser cavity. EDF: Erbium-doped fiber; SMF: Single-mode fiber; DCF: Dispersion compensation fiber; PC: Polarization controller; ISO: Polarization independent isolator.

3. Experimental observations

3.1. Net anomalous cavity dispersion regime

We first operate the laser in the net anomalous cavity dispersion regime by constructing the fiber ring laser with 3 m EDF, 13.4 m SMF and 12.6 m DCF. The average net cavity GVD is $-2.06 \text{ ps}^2/\text{km}$. Initially, the laser emits continuous waves (CWs) along the two orthogonal polarization directions. As we increase the pump power to 30dBm, polarization domains suddenly formed as a result of the strong incoherent cross coupling between the beams, as shown in Fig. 2(a). We note that the strength of the polarized light along each of the two orthogonal polarization directions is very different. Specifically, light polarized along the horizontal axis is much stronger than that along the vertical axis. This is also reflected by the 10 dB difference in their spectral intensity, shown in Fig. 2(c). Due to its strong intensity, a kind of slow frequency intensity modulation [36] starts to occur on the light. Under effect of nonlinear pulse shaping the intensity modulation is quickly evolved into a stable train of bright solitons. Once the bright solitons are formed, they start to move in the cavity. The cavity roundtrip time of our fiber laser is $\sim 145 \text{ ns}$. Several polarization domains with different domain sizes are formed in the cavity. The formed polarization domains are very stable so their domain walls are stationary in the cavity. Therefore, the bright solitons move from one domain to the another and constantly collide with the domain walls. We note that if one focuses only on the polarization domains in one polarization direction, each of these domains could also be considered as a bound state of the kink-anti-kink solitons [37]. The experimental result then also shows collisions of bright solitons with the stationary kink and anti-kink solitons. Our experimental result clearly shows that whenever the bright solitons meet a domain wall, they just simply pass through it. The collisions neither cause any visible changes of the bright solitons nor the domain walls, indicating that they collide elastically. As the formed bright solitons have a high peak intensity, due to the incoherent cross polarization coupling, each of them also induces a weak pulse on the orthogonal polarization direction. An intriguing feature

observed is that while in one type of polarization domains, or more specifically in the domains where the laser emission along the polarization is off, a weak bright pulse is induced, while in the other domains where the laser emission along the polarization is on, a weak dark pulse is induced, and the induced weak pulses move together with the inducing bright solitons, as shown in Fig. 2(b) and Visualization 1.

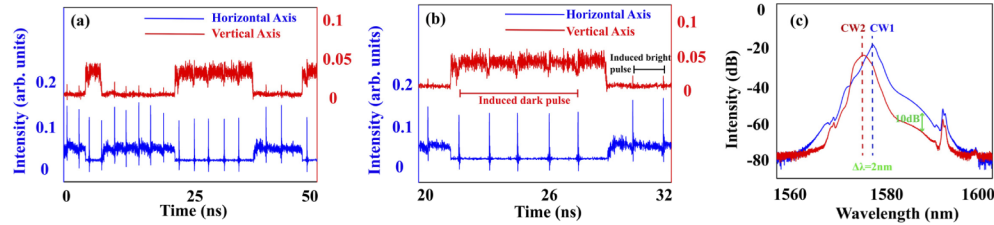


Fig. 2. (a) Polarization resolved laser emissions that show coexistence of bright solitons and polarization domains. (b) Zoom in of (a); (c) the corresponding polarization resolved optical spectra of (a).

Experimentally, by tuning the intracavity PC paddles, both the net cavity birefringence and the lasing intensity strength along the two orthogonal polarization directions could be altered. At a fixed pump power of 30 dBm, experimentally by carefully adjusting the PC paddles, a laser emission state with comparable lasing intensity along the two orthogonal polarization directions could be obtained. Under suitable conditions, coexistence of phase-locked vector bright solitons and polarization domains could be further achieved, as shown in Fig. 3(a). Different from the result shown in Fig. 2, where a train of scalar bright solitons are moving in the cavity, Fig. 3(a) shows now that a train of bright vector solitons are moving in the cavity and collide with the polarization domain walls. Our experimental result confirms that the collisions of the vector bright solitons with the polarization domain walls are elastic. For the purpose of making a direct comparison with a result obtained in the net normal cavity dispersion case, which will be discussed later, we have also measured the total laser output of the state, as shown in Fig. 3(b). In Fig. 3(b) the dark polarization domain wall soliton feature becomes clearly visible. Figure 3(c) is the polarization resolved optical spectra of the laser emissions. In the spectra both the Kelly sidebands [38] and the four-wave mixing (FWM) sidebands [39] are clearly visible, confirming that the vector bright solitons are formed by the coherent cross-polarization coupling between the two orthogonally polarized light beams. Therefore, Fig. 3 also shows the experimental evidence of coexistence of the phase locked vector bright solitons with the vector dark polarization domain wall solitons and their elastic collisions.

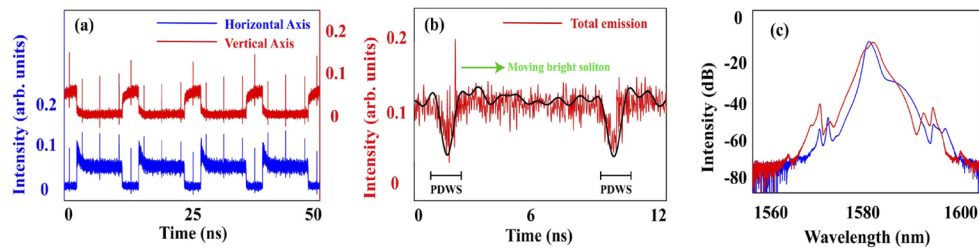


Fig. 3. (a) Polarization resolved laser emissions that show coexistence of phase-locked bright vector solitons and polarization domains; (b) Total laser emission where dark polarization domain wall solitons become visible; (c) Polarization resolved optical spectra of the laser emission.

In addition of elastic collisions, experimentally inelastic collisions between solitons and kink solitons were also revealed. Figure 4(a) shows a result obtained at a pump power of 22dBm, where both bright solitons and polarization domains are simultaneously formed in the laser. Figure 4(b) is a zoom-in of Fig. 4(a). To show the dynamics of the process, a short video record of the state is also displayed in [Visualization 2](#). Different from the case shown in Fig. 2 and Fig. 3, despite of the fact that bright solitons are formed and move in the domain, once they meet the domain wall, they then become disappeared, suggesting that the collisions are inelastic. Apart from the inelastic collision between vector bright solitons and the domain walls, we also observed the inelastic collisions between dark-bright vector solitons and the domain walls, as shown in Fig. 4(c) and [Visualization 3](#). Figure 4(d) is a zoom in of 4(c). Experimentally, whether an elastic collision or inelastic collision would occur depends on the relative strength between the polarization domains and the solitons. Specifically, when the strength of solitons is weaker than that of the polarization domains, the collision is inelastic, and the solitons will be trapped by the domain walls, on the contrast, if the soliton strength are comparable or stronger than that of the polarization domains, the collisions are elastic. In the elastic collision case solitons simply move across the polarization domains as shown in Fig. 2 and Fig. 3, which accords well with the numerical analysis shown in [40].

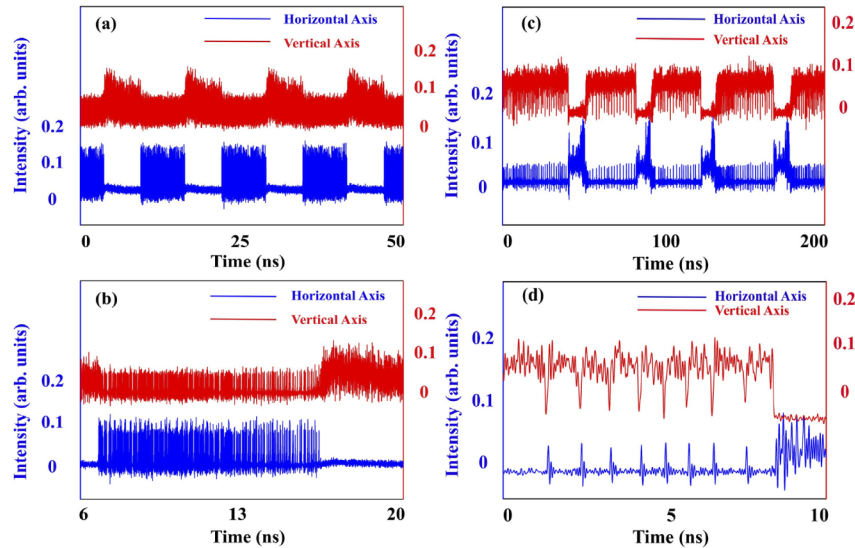


Fig. 4. (a) Bright vector solitons trapped by polarization domain walls; (b) Zoom in of (a); (c) Dark-bright vector solitons trapped by polarization domain walls; (d) Zoom in of (c).

3.2. Net normal cavity dispersion regime

We then operate the laser in the net normal cavity dispersion regime by constructing the fiber ring cavity with 3 m EDF, 5 m SMF and 20 m DCF, the averaged net cavity GVD is $6.1 \text{ ps}^2/\text{km}$. Although theoretically it has been shown that in the normal dispersion regime in the framework of coupled NLSEs, no polarization domains can be formed between two orthogonal linear polarization directions, it was first predicted by Haelterman and Sheppard that under coherent cross-polarization coupling, either a kind of dark-bright or dark-dark PDWS could be formed [23]. Indeed, experimentally, by carefully selecting the intracavity components and fine tuning the PC paddles, one could make the net cavity birefringence sufficiently small, and eventually achieve the coherent cross-polarization coupling between the two orthogonal polarization components of the laser emission. Under the laser operation, a state as shown in Fig. 5 is obtained. Figure 5(a)

shows the polarization resolved emissions of the laser and Fig. 5(d) are their corresponding optical spectra. The central wavelengths of the two orthogonally polarized laser emissions are well overlapped, confirming that they are coherently coupled. As predicted theoretically, the polarization resolved laser emissions exhibit the dark-bright PDWS features. Specifically, the observed PDWS have a pulse width in the order of several nanoseconds. However, in addition of the PDWS, scalar gray solitons with a pulse width in the order of tens of picoseconds are also formed simultaneously in the CW beam polarized along the vertical axis. With the term “gray solitons” we mean the dark solitons whose intensity dip does not drop to the zero intensity. As gray soliton formation has no threshold, they could be easily formed in fiber lasers with net normal cavity dispersion [41,42]. As shown in Fig. 5(a), multiple gray scalar solitons have been formed in the laser. Figure 5(b) further shows the dark-bright PDWS detected with a 1 GHz bandwidth detection system. In this case the narrow gray solitons are no longer detectable. The formed polarization domain wall solitons are stable and stationary in the cavity, while the formed gray solitons move with different velocities in the cavity. Figure 5(c) shows a measurement on the total laser emission of the same state. The situation shown in Fig. 5(c) is comparable to that shown in Fig. 3(b). It shows the coexistence of polarization domain wall solitons and scalar dark solitons in a fiber laser, and the scalar dark solitons collide with each other and with the dark domain wall solitons. However, different from the case obtained in the anomalous dispersion regime where the bright solitons maintain their pulse shape after collision, the gray solitons on one side move into the dark domain wall solitons, and on the other side, new gray solitons are continuously generated and move away from the domain wall dark soliton, indicating that the collisions could be inelastic in nature.

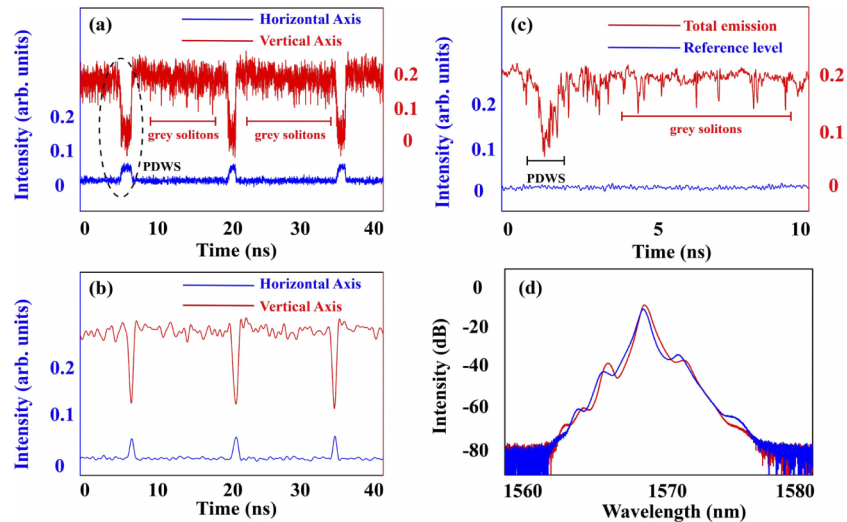


Fig. 5. (a) Coexistence of dark-bright PDWS and scalar gray solitons; (b) Detect with 1 GHz bandwidth; (c) Total laser emission; (d) Corresponding optical spectrum.

4. Discussion

Although the different types of solitons have different formation mechanisms, our experimental results show that once their formation conditions are fulfilled, they can be formed simultaneously in the same laser. The coexistence of different types of solitons in the same system allowed us to experimentally investigate the features of their interaction, especially, the collision between them. Operating the laser at net anomalous dispersion regime, we have experimentally confirmed

the existence of both elastic and inelastic collisions between the bright solitons with the kink, anti-kink, and polarization domain wall solitons. It is to emphasize that domain formation is a general feature in a fiber laser either with multiple wavelengths or orthogonally polarized emissions [14,15,24–28]. Our experimental results clearly show that in different domains the laser emission could exhibit different features. Based on this one could also explain why frequently in the same fiber laser one could even observe solitons with different features in different cavity sections. Obviously, the occurrence of optical domains in a fiber laser further complicates the solitons dynamics in a fiber laser. Therefore, special care has to be taken to distinguish the different conditions of soliton interaction in different domains.

In addition of the complication caused by the optical domains, it is also to note that different nonlinear effects, which could result in the formation of different types of solitons, could also coexist in a fiber laser under the same laser operation condition. As shown in our experiments, either the scalar or vector bright solitons could coexist with the domain wall solitons in the anomalous cavity dispersion regime, and scalar dark solitons coexist with the PDWS in the normal cavity dispersion regime. In previous experiments we have also observed the coexistence of both scalar and vector solitons in the same laser cavity [43], as well as coexistence of solitons with different central wavelengths in a fiber laser [44]. These experimental results show on one side the complicity of the real soliton fiber laser systems, on the other side, the opportunity to experimentally investigate the interactions among the formed solitons.

5. Conclusion

In conclusion, we have reported the first experimental observation on the coexistence of either bright or dark solitons with polarization domain wall solitons in a single mode fiber laser. Both elastic and inelastic collisions among different types of solitons are experimentally revealed. Specifically, in a net anomalous dispersion cavity fiber laser we have observed both the elastic and inelastic collisions of the bright solitons with the kink and anti-kink solitons, while in a net normal dispersion cavity fiber laser, the inelastic collision between scalar gray solitons and the dark polarization domain wall solitons. Our experimental results once again show that a nonlinear cavity fiber laser is a useful testbed for the experimental investigation on the various types of solitons and their interactions.

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References

1. A. Chabchoub, O. Kimmoun, H. Branger, N. Hoffmann, D. Proment, M. Onorato, and N. Akhmediev, "Experimental observation of dark solitons on the surface of water," *Phys. Rev. Lett.* **110**(12), 124101 (2013).
2. C. Chamon, "Solitons in carbon nanotubes," *Phys. Rev. B* **62**(4), 2806–2812 (2000).
3. K. E. Strecker, G. B. Partridge, A. G. Truscott, and R. G. Hulet, "Bright matter wave solitons in Bose-Einstein condensates," *New J. Phys.* **5**, 73 (2003).
4. Y. S. Kivshar and B. Luther-Davies, "Dark optical solitons: physics and applications," *Phys. Rep.* **298**(2-3), 81–197 (1998).
5. G. P. Agrawal, *Nonlinear Fiber Optics* (Academic, 2007).
6. L. F. Mollenauer, R. H. Stolen, and J. P. Gordon, "Experimental observation of picosecond pulse narrowing and solitons in optical fibers," *Phys. Rev. Lett.* **45**(13), 1095–1098 (1980).
7. P. Emplit, J. P. Hamaide, F. Reynaud, C. Forehly, and A. Barthelemy, "Picoscond steps and dark pulses through nonlinear single mode fibers," *Opt. Commun.* **62**(6), 374–379 (1987).
8. A. Mecozzi, S. Trillo, S. Wabnitz, and B. Daino, "All-optical switching and intensity discrimination by polarization modulation instability in periodically twisted fiber filters," *Opt. Lett.* **12**(4), 275–277 (1987).
9. S. G. Murdoch, R. Leonhardt, and J. D. Harvey, "Polarization modulation instability in weakly birefringent fibers," *Opt. Lett.* **20**(8), 866–868 (1995).
10. C. R. Menyuk, "Nonlinear pulse propagation in birefringent optical fibers," *IEEE J. Quantum Electron.* **23**(2), 174–176 (1987).

11. C. R. Menyuk, "Stability of solitons in birefringent optical fibers. I: Equal propagation amplitudes," *Opt. Lett.* **12**(8), 614 (1987).
12. M. Lisak, A. Hook, and D. Anderson, "Symbiotic solitary-wave pairs sustained by cross-phase modulation in optical fibers," *J. Opt. Soc. Am. B* **7**(5), 810–814 (1990).
13. D. N. Christodoulides, "Black and white vector solitons in weakly birefringent optical fibers," *Phys. Lett. A* **132**(8-9), 451–452 (1988).
14. C. Lecaplain, P. Grelu, and S. Wabnitz, "Polarization-domain-wall complexes in fiber lasers," *J. Opt. Soc. Am. B* **30**(1), 211–218 (2013).
15. D. Y. Tang, Y. F. Song, J. Guo, Y. J. Xiang, and D. Y. Shen, "Polarization domain formation and domain dynamics in a quasi-isotropic cavity fiber laser," *IEEE J. Select. Topics Quantum Electron.* **20**(5), 42–50 (2014).
16. B. A. Malomed, "Optical domain walls," *Phys. Rev. E* **50**(2), 1565–1571 (1994).
17. X. Hu, J. Guo, L. M. Zhao, J. Ma, and D. Y. Tang, "Coherently coupled vector black solitons in a quasi-isotropic cavity fiber laser," *Opt. Lett.* **45**(23), 6563–6566 (2020).
18. G. D. Shao, J. Guo, X. Hu, Y. F. Song, L. M. Zhao, and D. Y. Tang, "Vector dark solitons in a single mode fibre laser," *Laser Phys. Lett.* **16**(8), 085110 (2019).
19. J. Ma, G. D. Shao, Y. F. Song, L. M. Zhao, Y. J. Xiang, D. Y. Shen, M. Richardson, and D. Y. Tang, "Observation of dark-bright vector solitons in fiber lasers," *Opt. Lett.* **44**(9), 2185–2188 (2019).
20. X. Hu, J. Guo, G. D. Shao, Y. F. Song, S. W. Yoo, B. A. Malomed, and D. Y. Tang, "Observation of incoherently coupled dark-bright vector solitons in single-mode fibers," *Opt. Express* **27**(13), 18311–18317 (2019).
21. S. Wabnitz and B. Daino, "Polarization domains and instabilities in nonlinear optical fibers," *Phys. Lett. A* **182**(2-3), 289–293 (1993).
22. S. Pitotis, G. Millot, and S. Wabnitz, "Polarization domain wall solitons with counterpropagating laser beams," *Phys. Rev. Lett.* **81**(7), 1409–1412 (1998).
23. M. Haelterman and A. P. Sheppard, "Polarization domain walls in diffractive or dispersive Kerr media," *Opt. Lett.* **19**(2), 96–98 (1994).
24. H. Zhang, D. Y. Tang, L. M. Zhao, and R. J. Knize, "Vector domain wall solitons in a fiber ring laser," *Opt. Express* **18**(5), 4428–4433 (2010).
25. H. Zhang, D. Y. Tang, L. M. Zhao, and X. Wu, "Observation of polarization domain wall solitons in weakly birefringent cavity fiber lasers," *Phys. Rev. B* **80**(5), 052302 (2009).
26. Y. C. Meng, D. M. Zhang, G. Semaan, M. Kemel, A. Nady, M. Salhi, A. Komarov, and F. Sanchez, "Optical domains in fiber lasers," *J. Opt.* **23**(3), 035502 (2021).
27. H. Zhang, D. Y. Tang, L. M. Zhao, and X. Wu, "Dual-wavelength domain wall solitons in a fiber ring laser," *Opt. Express* **19**(4), 3525–3530 (2011).
28. Y. C. Meng, G. Semaan, M. Kemel, M. Salhi, A. Komarov, and F. Sanchez, "Color domain in fiber lasers," *Opt. Lett.* **43**(20), 5054–5057 (2018).
29. C. Anastassiou, M. Segev, K. Steiglitz, J. A. Giordmaine, M. Mitchell, M. F. Shih, S. Lan, and J. Martin, "Energy-Exchange interactions between Colliding Vector Solitons," *Phys. Rev. Lett.* **83**(12), 2332–2335 (1999).
30. J. S. Aitchison, A. M. Weiner, Y. Silberberg, M. K. Oliver, J. L. Jackel, D. E. Leaird, E. M. Vogel, and P. W. E. Smith, "Observation of spatial optical solitons in a nonlinear glass waveguide," *Opt. Lett.* **15**(9), 471–472 (1990).
31. J. P. Gordon, "Interaction forces among solitons in optical fibers," *Opt. Lett.* **8**(11), 596–598 (1983).
32. F. M. Mitschke and L. F. Mollenauer, "Experimental observation of interaction forces between solitons in optical fibers," *Opt. Lett.* **12**(5), 355–357 (1987).
33. R. Radhakrishnan, M. Lakshmanan, and J. Hietarinta, "Inelastic collision and switching of coupled bright solitons in optical fibers," *Phys. Rev. E* **56**(2), 2213–2216 (1997).
34. R. H. Goodman and R. Haberman, "Vector-soliton collision dynamics in nonlinear optical fibers," *Phys. Rev. E* **71**(5), 056605 (2005).
35. D. Rand, I. Glensk, C. S. Bres, D. A. Nolan, X. Chen, J. Koh, J. W. Fleischer, K. Steiglitz, and P. R. Prucnal, "Observation of temporal vector soliton propagation and collision in birefringent fiber," *Phys. Rev. Lett.* **98**(5), 053902 (2007).
36. D. Y. Tang, J. Guo, Y. F. Song, G. D. Shao, L. M. Zhao, and D. Y. Shen, "Temporal cavity soliton formation in an anomalous dispersion cavity fiber laser," *J. Opt. Soc. Am. B* **31**(12), 3050–3056 (2014).
37. W. Kim and H. T. Moon, "Dark and bright soliton exchange in a nonlinear dispersive medium," *J. Korean Phys. Soc.* **38**(5), 558–561 (2001).
38. S. M. J. Kelly, "Characteristic sideband instability of periodically amplified average soliton," *Electron. Lett.* **28**(8), 806 (1992).
39. H. Zhang, D. Y. Tang, L. M. Zhao, and N. Xiang, "Coherent energy exchange between components of a vector soliton in fiber lasers," *Opt. Express* **16**(17), 12618–12623 (2008).
40. X. D. Cao and B. A. Malomed, "Soliton-defect collisions in the nonlinear Schrödinger equation," *Phys. Lett. A* **206**(3-4), 177–182 (1995).
41. S. A. Gredeskul and Y. S. Kivshar, "Generation of dark solitons in optical fibers," *Phys. Rev. Lett.* **62**(8), 977 (1989).
42. D. Y. Tang, J. Guo, Y. F. Song, H. Zhang, L. M. Zhao, and D. Y. Shen, "Dark soliton fiber laser," *Opt. Express* **22**(16), 19831 (2014).

43. Y. Xiang, Y. Y. Luo, B. W. Liu, R. Xia, P. P. Shum, X. H. Tang, D. M. Liu, and Q. Z. Sun, "Scalar and Vector Solitons in a Bidirectional Mode-Locked Fibre Laser," *J. Lightwave Technol.* **37**(19), 5108–5114 (2019).
44. Y. F. Song, G. D. Shao, L. M. Zhao, D. Y. Shen, H. Zhang, and D. Y. Tang, "Dual-wavelength dissipative solitons in an anomalous-dispersion-cavity fiber laser," *Nanophotonics* **9**(8), 2361–2366 (2019).